

Teja Pavan Sai Singampalli

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EDUCATION

New Jersey Institute of Technology <i>Master of Science in Data Science (GPA: 3.92/ 4.00)</i>	Expected Graduation: May 2025 Newark, NJ
Indian Institute of Technology, Madras <i>Bachelor of Technology in Aerospace Engineering</i>	July 2017 – July 2021 Chennai, India

EXPERIENCE

Research Assistant <i>New Jersey Institute of Technology Dr. Jason T. L. Wang</i>	Nov 2024 – Present Newark, USA
- Applied SHAP and LIME for explainability in deep learning-based time series forecasting models - Conducted feature importance analysis to interpret predictions and identify key drivers influencing outputs - Developed visualizations (decision, summary, dependence plots) to explain time series model predictions	
Software Engineer <i>Societe Generale Global Solution Centre Java, Spring Boot, PostgreSQL</i>	July 2021 – Dec. 2023 Bangalore, India
- Boosted KYC process efficiency by 30% through automation, leveraging data analysis and problem-solving skills - Implemented 100+ robust APIs, enhancing data accuracy and saving 2 weeks across multiple business lines - Recognized with the prestigious SPOT Award by Manager as team MVP for exceptional performance	
Research Intern <i>Tata Consultancy Services</i>	May. 2020 – July 2020 Chennai, India
- Achieved 90% accuracy in converting natural language to SBVR and SQL models using NLP libraries - Produced 100+ SBVR-based rules and UML diagrams, improving business clarity by 70% for users	
Full-Stack Developer Intern <i>Indian Space Research Organisation (NESAC)</i>	Dec. 2019 – Jan. 2020 Umiam, India
- Established a Django site for 500+ image uploads, achieving upload times under 3 seconds - Enhanced backend via OpenCV for fast face detection, enriching visuals and user experience	

PROJECTS

Deep Learning-Based Speaker Recognition System <i>PyTorch, Transformer</i>	Oct. 2024 – Nov. 2024
- Designed a PyTorch speaker classifier using Transformer, reaching 96.25% accuracy on 600 unique speakers - Developed a data pipeline for 62,000+ audio features with mel-spectrogram extraction and batch processing - Constructed custom TransformerEncoder with self-attention and cosine learning rate scheduling for performance - Integrated model checkpointing to ensure consistent training and optimize model results	
Sequence-to-Sequence Neural Machine Translation <i>PyTorch, fairseq, NLP</i>	Sep. 2024 – Nov. 2024
- Developed an RNN-based English-to-Chinese translation model with PyTorch and fairseq - Applied subword tokenization, attention, and learning rate scheduling for improved translation accuracy - Attained a BLEU score of 15 on TED2020, highlighting deep learning proficiency - Optimizing model with Transformer trials and hyperparameter tuning to further increase BLEU score	
Matrix Predictor <i>Random Forest, Flask, React</i>	May 2024 – June 2024
- Engineered a robust random forest model accurately predicting 5x5 matrices with 21 green and 4 red cells - Implemented a Flask-React app for interactive selection of rows, columns, and matrices, achieving 84% accuracy	

TECHNICAL SKILLS

Languages: Java, Python, C, SQL (Postgres), HTML/CSS, R
Developer Tools: Git, Docker, Kubernetes, Jenkins, VS Code, PyCharm, IntelliJ, Eclipse
Libraries: pandas, NumPy, Matplotlib, Seaborn, Statistics, SciPy, Transformers, PyTorch
Concepts: Machine Learning, Deep Learning, Neural Networks, Artificial Intelligence, Explainable AI, Time Series, APIs, Hypothesis Testing, Data Preprocessing, Data Analysis, Data Visualization, Probability and Statistics